GPT-5.4 Solutions
Petropolitan Science (Re)Search 2026

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1 Problem 1, Petropolitan 2026

1.1 Variant 1 (33.0s)

We seek all pairs $(k, a) \in \mathbb{N} \times \mathbb{R}$ for which there exists a continuous $f : [0, 1] \to [0, 1]$ satisfying

$$\sum_{i=0}^{k-1} f_i(x) = a \qquad (\forall x \in [0, 1]),$$

where $f_0(x) = x$ and $f_n = f \circ f_{n-1}$.

We will prove that this is possible **if and only if**

$$a = \frac{k-1}{2}.$$

Moreover, for every k, such a function exists; for example $f(x) = 1 - x$ works when k is even, and $f(x) = \frac{1}{2}$ works when k is odd.

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1. Reformulation

Define

$$S(x) := \sum_{i=0}^{k-1} f_i(x).$$

The condition is that $S(x) \equiv a$ is constant on $[0, 1]$.

We must determine for which a this can happen.

---

2. Immediate examples

Before proving necessity, let us note two constructions.

Case 1: k even

Take

$$f(x) = 1 - x.$$

Then

$$f_0(x) = x, \quad f_1(x) = 1 - x, \quad f_2(x) = x, \quad f_3(x) = 1 - x, \ldots$$

Hence the sum of any two consecutive iterates is $x + (1 - x) = 1$. Since k is even,

$$S(x) = \frac{k}{2} \cdot 1 = \frac{k}{2}.$$

So in this case $a = \frac{k}{2}$.

But note that when k is even,

$$\frac{k-1}{2} \neq \frac{k}{2},$$

so this seems contradictory to the claimed answer. Therefore we must be careful: the problem states the sum runs from f_0 to f_{k-1}, so for $k = 2$, indeed $x + f(x) = 1$, giving $a = 1$. That suggests the true answer is not yet identified. So we continue carefully.

Case 2: k odd

Take

$$f(x) = \frac{1}{2}.$$

Then

$$f_0(x) = x, \qquad f_i(x) = \frac{1}{2} \quad (i \geq 1),$$

so

$$S(x) = x + \frac{k-1}{2},$$

which is not constant. So this does **not** work.

So we need a real derivation.

---

3. A key identity

Since $S(x) = a$ for all x, substituting $f(x)$ for x gives

$$\sum_{i=0}^{k-1} f_i(f(x)) = a.$$

But $f_i(f(x)) = f_{i+1}(x)$, so

$$\sum_{i=1}^{k} f_i(x) = a.$$

Subtracting from

$$\sum_{i=0}^{k-1} f_i(x) = a$$

yields

$$f_k(x) = f_0(x) = x \qquad (\forall x \in [0,1]).$$

Thus

$$f^k = \mathrm{id}_{[0,1]}\,.$$

So any admissible f is a continuous self-map of $[0,1]$ whose k-th iterate is the identity.

---

4. Continuous interval maps with a finite iterate equal to the identity

We now show that such an f must satisfy

$$f^2 = \mathrm{id}\,.$$

In fact, more strongly, f is either the identity or a decreasing involution.

Lemma

If $f : [0,1] \to [0,1]$ is continuous and $f^k = \mathrm{id}$ for some $k \geq 1$, then f is injective.

Proof. If $f(u) = f(v)$, then applying f^{k-1} to both sides gives

$$u = f^k(u) = f^{k-1}(f(u)) = f^{k-1}(f(v)) = f^k(v) = v.$$

So $u = v$. $\square$

A continuous injective map on an interval is strictly monotone. Hence f is either strictly increasing or strictly decreasing.

Lemma

If f is strictly increasing and $f^k = \mathrm{id}$, then $f = \mathrm{id}$.

Proof. Suppose there exists x with $f(x) > x$. Since f is increasing,

$$x < f(x) < f^2(x) < \cdots < f^k(x) = x,$$

impossible.

Similarly, if $f(x) < x$, then

$$x > f(x) > f^2(x) > \cdots > f^k(x) = x,$$

impossible.

Hence $f(x) = x$ for all x. $\square$

Lemma

If f is strictly decreasing and $f^k = \mathrm{id}$, then $f^2 = \mathrm{id}$.

Proof. Since f is decreasing, f^2 is increasing. Also

$$(f^2)^m = f^{2m}.$$

Because $f^k = \mathrm{id}$, some positive iterate of f^2 is also the identity (for example $f^{2k} = \mathrm{id}$). Thus f^2 is an increasing continuous map with a positive iterate equal to the identity. By the previous lemma,

$$f^2 = \mathrm{id}\,.$$

$\square$

Therefore every admissible f is either

1. $f(x) = x$, or 2. a decreasing involution: $f^2(x) = x$.

---

5. Compute the constant sum in each case

We now use the original equation.

Case A: $f(x) = x$

Then every iterate is x, so

$$S(x) = kx.$$

This is constant only if $k = 0$, impossible in $\mathbb{N}$, so this case gives no solutions.

Thus f must be decreasing and satisfy $f^2 = \mathrm{id}$.

Case B: f decreasing and $f^2 = \mathrm{id}$

Then the iterates alternate:

$$f_0(x) = x, \quad f_1(x) = f(x), \quad f_2(x) = x, \quad f_3(x) = f(x), \dots$$

Hence

$$S(x) = \begin{cases} \frac{k}{2}\big(x + f(x)\big), & k \text{ even}, \\ \frac{k-1}{2}\big(x + f(x)\big) + x, & k \text{ odd}. \end{cases}$$

We need this to be constant.

---

6. Even k

If $k = 2m$, then

$$S(x) = m(x + f(x)).$$

So $x + f(x)$ must be constant:

$$x + f(x) = c.$$

Since $f : [0,1] \to [0,1]$, evaluating at $x = 0, 1$ and using that f is decreasing involution forces $f(x) = 1 - x$, hence $c = 1$. Indeed from $f(x) = c - x$ and $f([0,1]) \subseteq [0,1]$, we must have $c = 1$.

Thus

$$a = S(x) = m = \frac{k}{2}.$$

And this is attained by $f(x) = 1 - x$.

So for even k, the only possible value is

$$a = \frac{k}{2}.$$

---

7. Odd k

Let $k = 2m + 1$. Then

$$S(x) = m(x + f(x)) + x.$$

If this is constant, then

$$m(x + f(x)) + x = a.$$

Applying f instead of x, and using $f^2(x) = x$, we get

$$m(f(x) + x) + f(x) = a.$$

Subtracting the two equations:

$$x - f(x) = 0.$$

So $f(x) = x$ for all x, contradicting the decreasing case.

Therefore there are **no solutions for odd k**.

(Indeed the identity case also failed.)

---

8. Conclusion

The only possible k are even, and then necessarily

$$a = \frac{k}{2}.$$

Conversely, for every even k, $f(x) = 1 - x$ works.

So the complete set of pairs is

$$\boxed{\{(k, a) \in \mathbb{N} \times \mathbb{R} : \ k \text{ is even and } a = \tfrac{k}{2}\}.}$$

Final answer.

$$\boxed{\text{There exists such a continuous } f : [0, 1] \to [0, 1] \text{ iff } k \text{ is even and } a = \frac{k}{2}.}$$

One example is $f(x) = 1 - x$, for which

$$x + (1 - x) + x + (1 - x) + \cdots = \frac{k}{2}.$$

1.2 Variant 2 (28.6s)

We seek all pairs $(k, a) \in \mathbb{N} \times \mathbb{R}$ such that there exists a continuous $f : [0,1] \to [0,1]$ satisfying

$$x + f(x) + f_2(x) + \cdots + f_{k-1}(x) = a \qquad (\forall x \in [0,1]),$$

where $f_0(x) = x$ and $f_n = f \circ f_{n-1}$.

We will determine exactly all such pairs.

---

1. Reformulation

Define

$$S(x) := \sum_{i=0}^{k-1} f_i(x).$$

The condition is

$$S(x) = a \qquad (\forall x \in [0,1]).$$

Thus S is constant.

We must find all $k \in \mathbb{N}$ and $a \in \mathbb{R}$ for which this is possible.

---

2. First easy observations

Since each $f_i(x) \in [0,1]$, we have

$$0 \le S(x) \le k,$$

so necessarily

$$a \in [0, k].$$

Also, evaluating at $x = 0$ and $x = 1$,

$$a = f(0) + f_2(0) + \cdots + f_{k-1}(0),$$

$$a = 1 + f(1) + f_2(1) + \cdots + f_{k-1}(1).$$

Subtracting gives

$$1 + \sum_{i=1}^{k-1} \big(f_i(1) - f_i(0)\big) = 0.$$

This already suggests strong rigidity.

---

3. A key identity

From

$$x + f(x) + \cdots + f_{k-1}(x) = a,$$

apply it to $f(x)$:

$$f(x) + f_2(x) + \cdots + f_k(x) = a.$$

Subtracting,

$$f_k(x) = x \qquad (\forall x \in [0,1]).$$

So $f^k = \mathrm{id}$.

This is the central structural consequence.

---

4. Continuous self-maps of an interval with finite order

We now prove:

Lemma

If $f : [0,1] \to [0,1]$ is continuous and $f^k = \mathrm{id}$ for some $k \geq 1$, then $f = \mathrm{id}$.

Proof

Since $f^k = \mathrm{id}$, f is bijective. A continuous bijection from $[0,1]$ to itself is strictly monotone.

So f is either strictly increasing or strictly decreasing.

- If f is strictly increasing and $f \neq \mathrm{id}$, then there exists x_0 with $f(x_0) \neq x_0$. - If $f(x_0) > x_0$, then by monotonicity

$$x_0 < f(x_0) < f^2(x_0) < \cdots < f^k(x_0),$$

hence $x_0 < f^k(x_0)$, contradiction since $f^k(x_0) = x_0$. - If $f(x_0) < x_0$, similarly

$$x_0 > f(x_0) > f^2(x_0) > \cdots > f^k(x_0),$$

contradiction.

Therefore an increasing f with $f^k = \mathrm{id}$ must be $f = \mathrm{id}$.

- If f is strictly decreasing, then f^2 is strictly increasing. Also

$$(f^2)^k = f^{2k} = (f^k)^2 = \mathrm{id}.$$

By the increasing case, $f^2 = \mathrm{id}$.

But a strictly decreasing involution on $[0,1]$ certainly exists (for example $x \mapsto 1 - x$); however if $f^2 = \mathrm{id}$, then our original sum condition will force more. We postpone this for a moment.

Actually we can rule out the decreasing case directly using the constant-sum equation: if f is decreasing, then f^2 is increasing, and since $f^k = \mathrm{id}$, the parity matters: - if k is odd, f^k is decreasing, impossible because id is increasing; - if k is even, then f^k is increasing, possible in principle, but then $f^2 = \mathrm{id}$.

So the only nontrivial possibility from $f^k = \mathrm{id}$ alone is a decreasing involution when k is even. We will show next that the sum condition excludes it unless it coincides with the identity case $k = 1$, which it does not.

This completes the structural reduction. $\square$

---

5. Use the sum condition with $f^k = \mathrm{id}$

We now examine the possibilities.

Case 1: $f = \mathrm{id}$

Then

$$f_i(x) = x \quad \text{for all } i,$$

so

$$S(x) = kx.$$

This is constant on $[0,1]$ only if $k=0$, but $k \in \mathbb{N}$ and in the usual convention $k \geq 1$. For $k \geq 1$, kx is not constant.

Thus no solution arises from $f = \mathrm{id}$.

So if a solution exists, it would have to come from the decreasing involution possibility.

Case 2: f is decreasing and $f^2 = \mathrm{id}$

Then k must be even, say $k = 2m$, because $f^k = \mathrm{id}$.

Now the iterates alternate:

$$f_{2r}(x) = x, \qquad f_{2r+1}(x) = f(x).$$

Hence

$$S(x) = m\,x + m\,f(x) = m\big(x + f(x)\big).$$

So the condition $S(x) = a$ is equivalent to

$$x + f(x) = c := \frac{a}{m} \qquad (\forall x \in [0,1]).$$

Thus necessarily

$$f(x) = c - x.$$

Since $f : [0,1] \to [0,1]$, for all $x \in [0,1]$ we need $c - x \in [0,1]$. In particular: - at $x = 0$: $c \in [0,1]$, - at $x = 1$: $c - 1 \in [0,1]$, i.e. $c \in [1,2]$.

Therefore $c = 1$, and hence

$$f(x) = 1 - x.$$

Then

$$S(x) = m(x + 1 - x) = m.$$

So

$$a = m = \frac{k}{2}.$$

Thus for every even k, the function $f(x) = 1 - x$ gives a solution with

$$a = \frac{k}{2}.$$

---

6. Show there are no other solutions

We have already shown any solution must satisfy $f^k = \mathrm{id}$. For a continuous self-map of $[0,1]$, this implies f is a homeomorphism, hence monotone.

- If f is increasing, then $f = \mathrm{id}$, impossible. - Hence f is decreasing. Then k must be even, and $f^2 = \mathrm{id}$. - Therefore the iterates alternate, and the constant-sum condition gives $x + f(x) = \text{constant}$. - Since $f : [0,1] \to [0,1]$, this constant must be 1, so $f(x) = 1 - x$, and $a = k/2$.

So the above family is the only one.

---

7. Conclusion

The pairs (k, a) for which such a continuous f exists are exactly those with k even and

$$a = \frac{k}{2}.$$

Indeed, for any even k, the function

$$f(x) = 1 - x$$

satisfies

$$f_{2r}(x) = x, \qquad f_{2r+1}(x) = 1 - x,$$

hence

$$x + f(x) + \cdots + f_{k-1}(x) = \frac{k}{2}\,x + \frac{k}{2}\,(1 - x) = \frac{k}{2}.$$

Final answer.

$$\boxed{\text{Exactly all pairs } (k, a) \text{ with } k \text{ even and } a = \frac{k}{2}.}$$

A corresponding function is $\boxed{f(x) = 1 - x}$.

1.3 Variant 3 (17.7s)

We seek all pairs $(k, a) \in \mathbb{N} \times \mathbb{R}$ for which there exists a continuous $f : [0,1] \to [0,1]$ satisfying

$$\sum_{i=0}^{k-1} f_i(x) = a \qquad (\forall x \in [0,1]),$$

where $f_0(x) = x$ and $f_n = f \circ f_{n-1}$.

We will determine all such pairs completely.

---

1. Reformulation

Define

$$S_k(x) := x + f(x) + f_2(x) + \cdots + f_{k-1}(x).$$

We are given that

$$S_k(x) \equiv a \quad \text{on } [0,1].$$

We must find all $k \in \mathbb{N}$ and $a \in \mathbb{R}$ for which such a continuous f exists.

Since each term lies in $[0,1]$, clearly

$$0 \le a \le k.$$

But we need the exact possibilities.

---

2. First easy cases

Case $k = 1$

Then the condition is just

$$f_0(x) = x = a \qquad \forall x \in [0,1],$$

impossible since x is not constant.

So:

$$k \neq 1.$$

---

3. A key identity

Since

$$x + f(x) + \cdots + f_{k-1}(x) = a,$$

replacing x by $f(x)$ gives

$$f(x) + f_2(x) + \cdots + f_k(x) = a.$$

Subtracting,

$$f_k(x) = x \qquad \forall x \in [0,1].$$

So any such f must satisfy

$$f^k = \mathrm{id}_{[0,1]}.$$

This is very restrictive.

---

4. Continuous self-maps of an interval with finite order

We now prove:

Lemma

If $f : [0,1] \to [0,1]$ is continuous and $f^k = \mathrm{id}$ for some $k \geq 1$, then $f = \mathrm{id}$.

Proof Suppose not. Then there exists $x_0 \in [0,1]$ with $f(x_0) \neq x_0$.

If $f(x_0) > x_0$, then by continuity,

$$f^n(x_0) > f^{n-1}(x_0) > \cdots > f(x_0) > x_0$$

as long as equality never occurs earlier. In particular the orbit is strictly increasing, so $f^k(x_0) > x_0$, contradicting $f^k(x_0) = x_0$.

Similarly, if $f(x_0) < x_0$, then the orbit is strictly decreasing, hence $f^k(x_0) < x_0$, again a contradiction.

Therefore $f(x) = x$ for all x, so $f = \mathrm{id}$. $\square$

A slightly more conceptual argument is that a continuous injective map on an interval is monotone, and if $f^k = \mathrm{id}$, then f is injective; if increasing, it must be the identity, and if decreasing, then $f^2 = \mathrm{id}$, impossible on $[0,1]$ unless fixed everywhere. But the above direct proof suffices.

Hence from the functional equation we conclude:

$$f(x) = x \qquad \forall x \in [0,1].$$

---

5. Determine a

If $f(x) = x$, then

$$f_i(x) = x \quad \text{for all } i,$$

so

$$x + f(x) + \cdots + f_{k-1}(x) = kx.$$

For this to be constant in x, we would need $kx = a$ for all $x \in [0,1]$, impossible for every $k \in \mathbb{N}$.

Thus no such pair exists.

---

6. Check for any overlooked exceptions

- $k = 1$: impossible. - $k \geq 2$: the equation implies $f^k = \mathrm{id}$, hence $f = \mathrm{id}$, hence the sum is kx, not constant.

So indeed there are no solutions.

Final answer.

$$\boxed{\text{There are no pairs } (k,a) \in \mathbb{N} \times \mathbb{R} \text{ for which such a continuous } f \text{ exists.}}$$

1.4 Final Synthesis (15.1s, total 94.4s)

Among the drafts, **Solution 2 is essentially correct**. **Solution 1** has a false start but eventually reaches the right conclusion. **Solution 3** is incorrect: it wrongly claims that a continuous interval map with $f^k = \mathrm{id}$ must be the identity, overlooking decreasing involutions such as $f(x) = 1 - x$.

Here is a concise, corrected complete solution.

$$\text{Let } S(x) := \sum_{i=0}^{k-1} f_i(x), \qquad f_0(x) = x, \quad f_{n+1} = f \circ f_n.$$

We are given that

$$S(x) = a \qquad \forall x \in [0, 1].$$

We determine all possible pairs (k, a).

1. First consequence: $f^k = \mathrm{id}$

Apply the identity $S(x) = a$ at the point $f(x)$. Since $f_i(f(x)) = f_{i+1}(x)$, we get

$$\sum_{i=1}^{k} f_i(x) = a.$$

Subtracting from

$$\sum_{i=0}^{k-1} f_i(x) = a$$

yields

$$f_k(x) = x \qquad \forall x \in [0, 1].$$

Hence

$$f^k = \mathrm{id}_{[0,1]}.$$

2. Structure of such continuous maps

From $f^k = \mathrm{id}$, f is injective: if $f(u) = f(v)$, then applying f^{k-1} gives $u = v$. A continuous injective map on an interval is strictly monotone, so f is either strictly increasing or strictly decreasing.

If f is increasing

Assume $f(x_0) > x_0$ for some x_0. Then by monotonicity,

$$x_0 < f(x_0) < f^2(x_0) < \cdots < f^k(x_0) = x_0,$$

impossible. Similarly $f(x_0) < x_0$ is impossible. Thus $f(x) = x$ for all x, i.e. $f = \mathrm{id}$.

But then

$$S(x) = kx,$$

which is not constant on $[0, 1]$. So the increasing case gives no solution.

Therefore f is decreasing

Then f^2 is increasing, and some positive iterate of f^2 is the identity, namely

$$(f^2)^k = f^{2k} = \mathrm{id}.$$

By the increasing-case argument, this forces

$$f^2 = \mathrm{id}.$$

So f is a decreasing involution.

Also, since $f^k = \mathrm{id}$, k must be even: if k were odd, then f^k would be decreasing, but the identity is increasing.

Thus write

$$k = 2m.$$

3. Use the sum condition

Because $f^2 = \mathrm{id}$, the iterates alternate:

$$f_{2r}(x) = x, \qquad f_{2r+1}(x) = f(x).$$

Hence

$$S(x) = m\,x + m\,f(x) = m(x + f(x)).$$

Since $S(x)$ is constant, $x + f(x)$ must be constant:

$$x + f(x) = c \qquad \forall x \in [0,1].$$

Therefore

$$f(x) = c - x.$$

Now $f([0,1]) \subseteq [0,1]$. From $x = 0$ we get $c \in [0,1]$, and from $x = 1$ we get $c-1 \in [0,1]$, i.e. $c \in [1,2]$. Hence

$$c = 1.$$

So necessarily

$$f(x) = 1 - x.$$

Then

$$S(x) = m(x + 1 - x) = m = \frac{k}{2}.$$

Thus

$$a = \frac{k}{2}.$$

4. Existence

For every even k, the function

$$f(x) = 1 - x$$

is continuous from $[0,1]$ to $[0,1]$, and its iterates alternate:

$$x,\ 1-x,\ x,\ 1-x, \ldots$$

So for $k = 2m$,

$$\sum_{i=0}^{k-1} f_i(x) = m(x + 1 - x) = m = \frac{k}{2}.$$

Hence such a function exists.

Final answer.

$$\boxed{\text{Such a continuous } f : [0,1] \to [0,1] \text{ exists exactly when } k \text{ is even and } a = \frac{k}{2}.}$$

In that case one may take

$$\boxed{f(x) = 1 - x.}$$